Daniel Joseph

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Education

McMaster University Expected Apr 2030

B.Eng. & Management, Computer Engineering (Level II) *Hamilton, ON*

Relevant Coursework: Circuit Analysis; Circuits & Waves; Principles of Programming; Data Structures & Algorithms.

Projects

32-bit RV32I CPU Core — RTL Design & Verification

SystemVerilog, cocotb, Verilator, Python

- Designed a synthesizable **32-bit RV32I single-cycle CPU** with a **32×32-bit register file**, ALU, instruction decoder, immediate generator, branch/jump logic, and instruction/data-memory interfaces.
- Built self-checking **cocotb/Verilator** regressions and a reusable **Python instruction encoder** to generate `.hex` programs; checked register writeback, load/store behavior, branch targets, and jump links.
- Ran **FPGA-targeted synthesis and implementation**; analyzed utilization, timing, and critical-path reports to confirm synthesizability and identify the longest combinational path.

ESP32 Assistive Grabber — Embedded Systems & PCB Design

ENGINEER 1P13 Client-Centered Design Project

- Developed a wheelchair-mounted grabber for a client with arthritis, integrating an **ESP32**, custom **Altium PCB design**, motor actuation, and **C++ grasp-control firmware** into an electromechanical prototype.
- Iterated **3 full prototypes** and tested pickup time, load/torque, mass, ergonomics, and wheelchair fit against requirements; documented failure modes and design refinements.

Experience

Electrical Sub-Team Member Sept 2026 – Present

McMaster DARE — BELUGA Underwater Drone

- Developing **microcontroller subsystems for sensors** to support sensor interfacing and data acquisition for **BELUGA**, an underwater drone.
- Integrating **INA3221 current monitoring** for **I^2C**-based measurement of supply current and bus voltage within BELUGA's electrical system.

Technical Operations Intern Jun 2026 – Aug 2026

Acden Environment *Fort McMurray, AB*

- Built a **Python API pipeline** connecting TRUX logistics records with **Salesforce Agentforce** contract data, reducing manual reconciliation by **85%**; implemented state-machine error handling and recovery across both APIs.
- Calculated contract burn rates and hauling cycle times from dispatch data; serialized **JSON** payloads for SLA analysis to flag under-delivery risk up to **3 months in advance**.
- Automated purchase-order and PPE workflows with **Power Apps, SharePoint, and Power Automate** across **12 sites** and **3 approvers**; eliminated **2 manual touches** for eligible requests across **5 categories up to $1,000**.

Embedded Controls & Mechanical Design Lead Sept 2023 – May 2025

Westwood Robotics Program — VEX V5 Team 221X *Fort McMurray, AB*

- Developed **C++** autonomous and driver-control software; tuned positional and angular **PID controllers** with IMU/encoder sensor fusion, achieving **±3 cm** position and **±0.5°** heading repeatability across **200+ test runs**.
- Built a real-time optical-sensor color-sorting subsystem and reduced mechanical iteration time by **25%** through **SolidWorks** fit and clearance validation.
- Won **Alberta Provincials** with **Team 221X** and competed at the **2025 VEX Robotics World Championship** in **Dallas, Texas**.

Technical Skills

Digital Design: SystemVerilog, Verilog, RISC-V (RV32I), cocotb, Verilator, self-checking testbenches
Embedded: C++, Embedded C, ESP32, PID control, sensor fusion, IMU/encoders/optical sensors, motor actuation
Hardware & CAD: Hardware debugging, PCB design, Altium Designer, KiCad, SolidWorks, AutoCAD, 3D printing
Software & Tools: Python, REST APIs, JSON, UML, Git, Power Apps, SharePoint, Power Automate